

Mehmet Emin Copuroglu

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SKILLS

- Embedded: STM32, Arduino, Embedded C
- Hardware: PCB Design, Schematic Capture, Component Selection
- Tools: KiCad, STM32CubeIDE, Fusion 360, AutoCAD Electrical, LTSpice, Microsoft Offices
- Industrial: PLC Basics, Industrial Instrumentation, Process Control, Motor Protection, Sensors & Transmitters, KUKA Robots

WORK EXPERIENCE

Workshop Assistant, Teknik Masura – Mersin TR

July 2024 – August 2024

- Assisted mechanical, electrical, and maintenance teams in workshop and industrial operations, including basic machining, material preparation, welding, mechanical assembly, and electrical panel wiring/maintenance using industrial tools and equipment
- Supported hands-on production tasks by contributing to machining, assembly, and electrical work while gaining experience with workshop processes and equipment

Electrical Team Member, Queen's Knights Robotics Team – Kingston ON

September 2025 – Present

- Designed a turret control PCB as part of a team-based breakout board project
- Developed schematics, selected electronic components, and created PCB layouts using KiCad

Electrical Engineering Intern, Ankutsan – Adana TR

July 2026 – August 2026

- Assisted electrical engineers and maintenance technicians with preventive maintenance, troubleshooting, and daily operations in a highly automated manufacturing facility.
- Gained hands-on exposure to PLC-based automation, industrial instrumentation, three-phase motors, VFDs, electrical panels, sensors, transmitters, calibration, and 4–20 mA current loop systems.
- Created and maintained a comprehensive Excel-based motor inventory by documenting technical specifications and equipment data.
- Collaborated with engineers across multiple departments to develop a broader understanding of industrial production processes, equipment reliability, and maintenance practices.

PROJECTS

Automated Pre-Treatment Water System

September 2025 – December 2025

Collaborated in a five-member team to develop an automated pre-treatment water system. Developed Arduino-based embedded software for pump control, chemical dosing, sensor monitoring, and mixing sequences while contributing to electrical integration, system testing, and automation reliability improvements.

STM32 Button Interface Board

February 2026 – March 2026

Designed a custom STM32F303-based PCB in KiCad, including schematic capture, component selection, and PCB layout. Developed modular Embedded C firmware in STM32CubeIDE using GPIO, timers, interrupts, and software debounce. Published the complete hardware design, firmware, and documentation on GitHub.

Railway Track Condition Monitoring System

January 2026 – March 2026

Worked in a five-member team to develop an embedded railway track condition monitoring prototype. Integrated sensors with microcontroller-based data acquisition, performed hardware testing and system validation, and contributed to electrical integration, prototype optimization, and technical documentation for reliable condition-based maintenance.

Sound Lab – Experimental Browser Synth

April 2026 – Present

Developed an interactive browser-based synthesizer using JavaScript and the Web Audio API, implementing oscillators, filters, envelopes, and real-time audio effects. Designed waveform visualization, modular software architecture, and responsive controls while applying digital signal processing concepts to interactive audio synthesis.

EDUCATION

Queen's University

Kingston, ON

Electrical Engineering

2025-2029